

HR Employee Data Analysis

Employee Workforce, Turnover & Demographic Analysis

Tools

Excel | MySQL | Power BI

CREATED BY

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PROJECT OVERVIEW

This project analyzes HR employee data using SQL and Power BI. The objective is to clean raw employee data, answer key HR business questions using SQL, and develop an interactive dashboard to support workforce analysis and decision-making.

Tools Used

- MySQL
- Power BI
- Excel

Skills Demonstrated

- Data Cleaning
- SQL
- Data Analysis
- Data Visualization
- Dashboard Design
- KPI Reporting
- HR Analytics

Business Questions

- Gender breakdown
- Race distribution
- Age distribution
- Headquarters vs Remote
- Average employment length
- Gender by department
- Job title distribution
- Department turnover
- Employees by state
- Employee growth over time
- Department tenure

PROJECT WORKFLOW

Raw Dataset



Data Cleaning (SQL)



Data Analysis (SQL)



Power BI Dashboard



Business Insights

DATA & CLEANING

RAW DATA

	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	id	first_name	last_name	birthdate	gender	race	department	jobtitle	location	hire_date	termdate	location_cit	location_state	
2	00-0037846	Kimmy	Walczynski	6/04/1991	Male	Hispanic or	Engineering	Programme	Headquarte	1/20/2002		Cleveland	Ohio	
3	00-0041533	Ignatius	Springett	6/29/1984	Male	White	Business De	Business An	Headquarte	4/08/2019		Cleveland	Ohio	
4	00-0045747	Corbie	Bittlestone	7/29/1989	Male	Black or Afr	Sales	Solutions Er	Headquarte	10/12/2010		Cleveland	Ohio	
5	00-0055274	Baxy	Matton	9/14/1982	Female	White	Services	Service Tec	Headquarte	4/10/2005		Cleveland	Ohio	
6	00-0076100	Terrell	Suff	4/11/1994	Female	Two or Mor	Product Ma	Business An	Remote	9/29/2010	2029-10-29	Flint	Michigan	
7	00-0116166	Kacie	Offiler	1/18/1971	Male	Asian	Engineering	Developer I	Headquarte	9/01/2018		Cleveland	Ohio	
8	00-0363185	Sandro	Admans	11/19/1979	Male	Two or Mor	Product Ma	Quality Engi	Headquarte	11/08/2012		Cleveland	Ohio	
9	00-0380704	Eugene	Lehrahan	10/14/1988	Female	Black or Afr	Engineering	Developer I	Headquarte	6/27/2007		Cleveland	Ohio	
10	00-0381660	Wainwright	Corfield	12/13/1996	Male	Asian	Engineering	Business Sys	Headquarte	2/20/2001	2008-12-05	Cleveland	Ohio	
11	00-0419202	Dyann	Isoldi	3/27/1980	Male	Two or Mor	Engineering	Web Develc	Headquarte	1/27/2005		Cleveland	Ohio	
12	00-0472287	Grantley	Oret	9/06/1975	Male	Two or Mor	Services	Service Tec	Headquarte	11/01/2004		Cleveland	Ohio	
13	00-0472832	Elmore	Worner	1/07/1966	Female	White	Engineering	Business Sys	Headquarte	12/05/2000		Cleveland	Ohio	
14	00-0566380	Dud	Brain	3/17/1984	Male	Two or Mor	Business De	Business An	Headquarte	9/17/2008		Cleveland	Ohio	
15	00-0571075	Aguie	Conford	11/02/1971	Male	White	Business De	Research As	Headquarte	11/25/2015		Cleveland	Ohio	
16	00-0624189	Katerina	Rosborough	8/20/1967	Male	Hispanic or	Engineering	Analyst Prog	Headquarte	5/17/2019		Cleveland	Ohio	
17	00-0715212	Alida	Longley	1/28/1973	Female	American In	Accounting	Staff Accou	Headquarte	2/04/2002		Cleveland	Ohio	
18	00-0755645	Laraine	Petre	5/11/1967	Male	White	Engineering	Software Er	Headquarte	9/30/2010		Cleveland	Ohio	
19	00-0778934	Gareth	MacCook	2/21/1987	Female	Black or Afr	Legal	Senior Attor	Remote	2/18/2010		Flint	Michigan	
20	00-0794247	Scottie	Chestney	2/23/1972	Female	Black or Afr	Engineering	Data Visuali	Headquarte	7/03/2002		Cleveland	Ohio	
21	00-0948136	Christophor	Boseley	5/23/1983	Female	Two or Mor	Marketing	Senior Edito	Headquarte	9/06/2007		Cleveland	Ohio	
22	00-0971612	Arleyne	Froome	8/01/1999	Male	Two or Mor	Engineering	Software Cc	Headquarte	4/09/2015		Cleveland	Ohio	
23	00-1051096	Todd	Cashen	6/19/1999	Female	White	Accounting	Financial An	Remote	6/20/2014		Pittsburgh	Pennsylvania	
24	00-1052230	Elmo	McNee	12/16/1988	Female	Black or Afr	Services	Service Mar	Headquarte	7/18/2002	2006-05-22	Cleveland	Ohio	
25	00-1100714	Regen	Nafzger	4/06/1990	Male	White	Human Resc	Senior Recri	Headquarte	7/23/2016	2022-07-12	Cleveland	Ohio	
26	00-1147503	Penelope	Wenman	12/19/1994	Male	White	Business De	Research As	Headquarte	9/10/2019		Cleveland	Ohio	
27	00-1189819	Rudolf	Reichardt	6/05/1979	Male	Black or Afr	Business De	Business An	Remote	5/07/2007	2022-02-25	Philadelphia	Pennsylvania	

VIEW RAW DATA HERE

https://drive.google.com/file/d/1wBhJLIXBOovyjkjQdkNYtvEg4PAZ2HQb/view?usp=drive_link

DATA & CLEANING

TURNOVER RATE

	A	B	C	D	E
1	department	total_count	terminated_	termination_rate	
2	Auditing	50	10	0.2	
3	Legal	299	51	0.1706	
4	Training	1622	226	0.1393	
5	Support	903	122	0.1351	
6	Sales	1745	233	0.1335	
7	Human Resc	1727	227	0.1314	
8	Engineering	6387	836	0.1309	
9	Accounting	3192	412	0.1291	
10	Services	1618	206	0.1273	
11	Research ar	1032	130	0.126	
12	Product Ma	623	78	0.1252	
13	Business De	1569	181	0.1154	
14	Marketing	480	53	0.1104	

AGE GROUP_GENDER_COUNT

1	age_group	gender	count
2	18-24	Female	289
3	18-24	Male	295
4	18-24	Non-Confor	17
5	25-34	Female	2429
6	25-34	Male	2647
7	25-34	Non-Confor	132
8	35-44	Female	2444
9	35-44	Male	2766
10	35-44	Non-Confor	149
11	45-54	Female	2386
12	45-54	Male	2588

JOBTITLE_COUNT

1	jobtitle	count
2	Web Developer IV	60
3	Web Developer III	57
4	Web Developer II	67
5	Web Developer I	83
6	Web Designer IV	5
7	Web Designer III	11
8	Web Designer II	4
9	Web Designer I	28
10	VP Sales	5
11	VP Quality Control	26

Cleaning performed

- Standardized date formats
- Handled missing/invalid termination dates
- Identified invalid age values
- Filtered inappropriate employee records
- Prepared data for analysis

DATA CLEANING FILES →



SQL ANALYSIS

Average Employee Tenure

```
156 • SELECT
157     department,
158     ROUND(AVG(DATEDIFF(termdate, hire_date) / 365), 0) AS avg_tenure
159 FROM hr
160 WHERE termdate IS NOT NULL
161     AND termdate <= CURDATE()
162     AND age >= 18
163 GROUP BY department;
```

department	avg_tenure
Engineering	9
Services	9
Human Resources	8
Business Development	8
Sales	9
Support	8
Auditing	8
Training	8
Accounting	8
Research and Development	8
Product Management	8
Legal	8
Marketing	9

Employee Hiring & Termination Trend Analysis

```
131
132 • SELECT
133     year,
134     hires,
135     terminations,
136     hires - terminations AS net_change,
137     ROUND(((hires - terminations) / hires) * 100, 2) AS net_change_percent
138 FROM (
139     SELECT
140         YEAR(hire_date) AS year,
141         COUNT(*) AS hires,
142         SUM(
143             CASE
144                 WHEN termdate IS NOT NULL
145                     AND termdate <= CURDATE()
146                 THEN 1
147                 ELSE 0
148             END
149         ) AS terminations
150     FROM hr
151     WHERE age >= 18
152     GROUP BY YEAR(hire_date)
153 ) AS employee_trends;
```

year	hires	terminations	net_change	net_change_percent
2000	211	26	185	87.68
2001	1082	197	885	81.79
2002	1012	162	850	83.99
2003	1088	198	890	81.80

Employee Distribution by State

```
Query 1 x
125 • select location_state, count(*) as count
126 from hr
127 WHERE age >= 18
128     AND (termdate IS NULL OR termdate > CURDATE())
129 group by location_state
130 order by count desc;
131
```

location_state	count
Ohio	14980
Pennsylvania	937
Illinois	734
Indiana	576
Michigan	573
Kentucky	362
Wisconsin	317

View Complete SQL Cleaning Script :



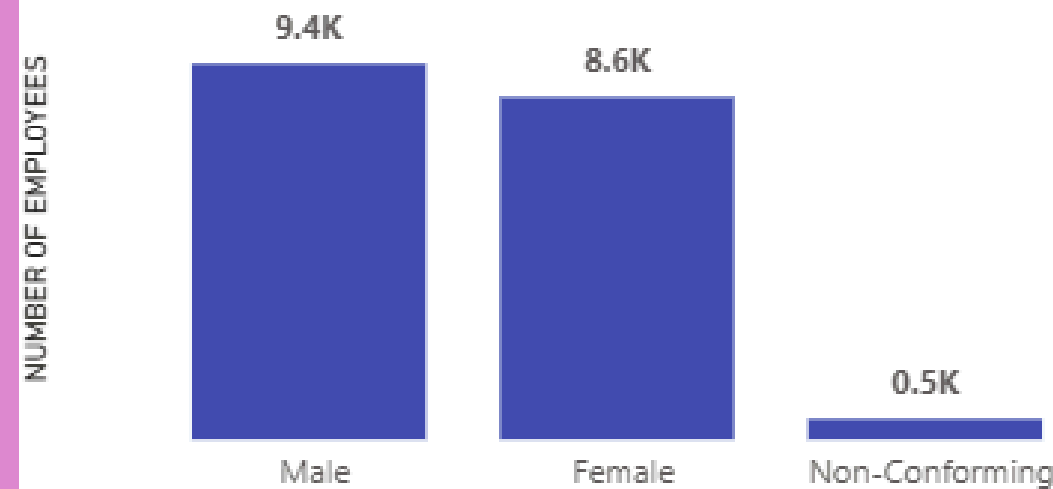
POWER BI DASHBOARD

HR EMPLOYEE DISTRIBUTION REPORT

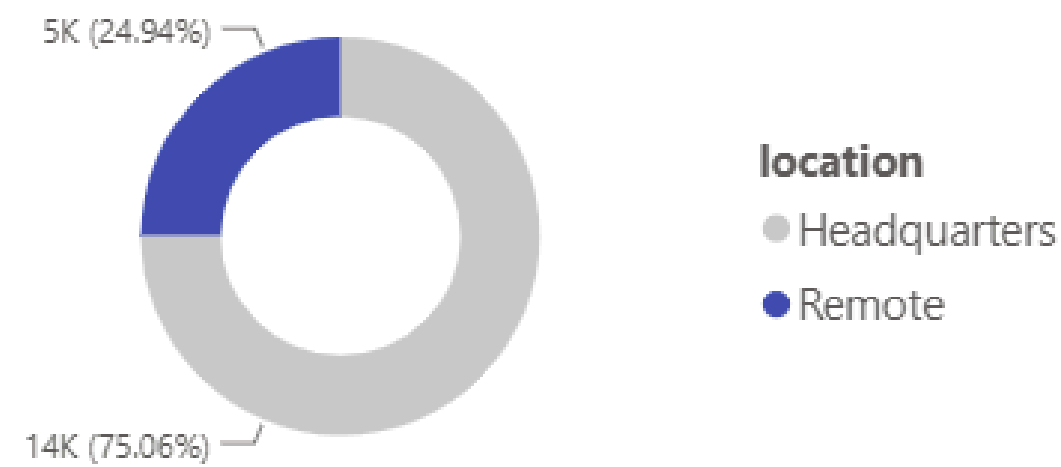
AVERAGE LENGTH OF
EMPLOYMENT (YEARS)

8

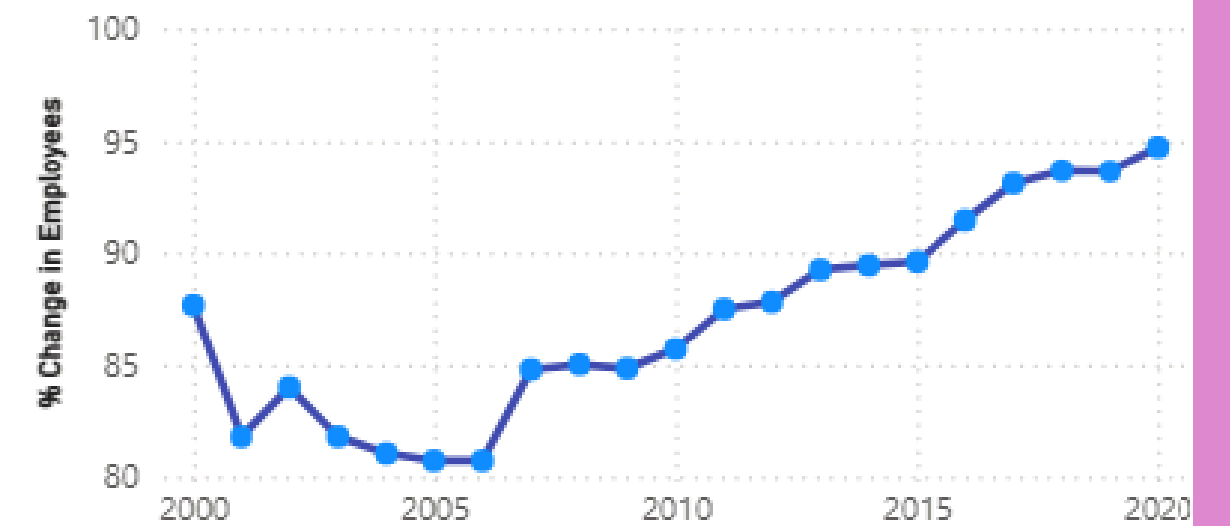
Gender Distribution



headquarters vs remote



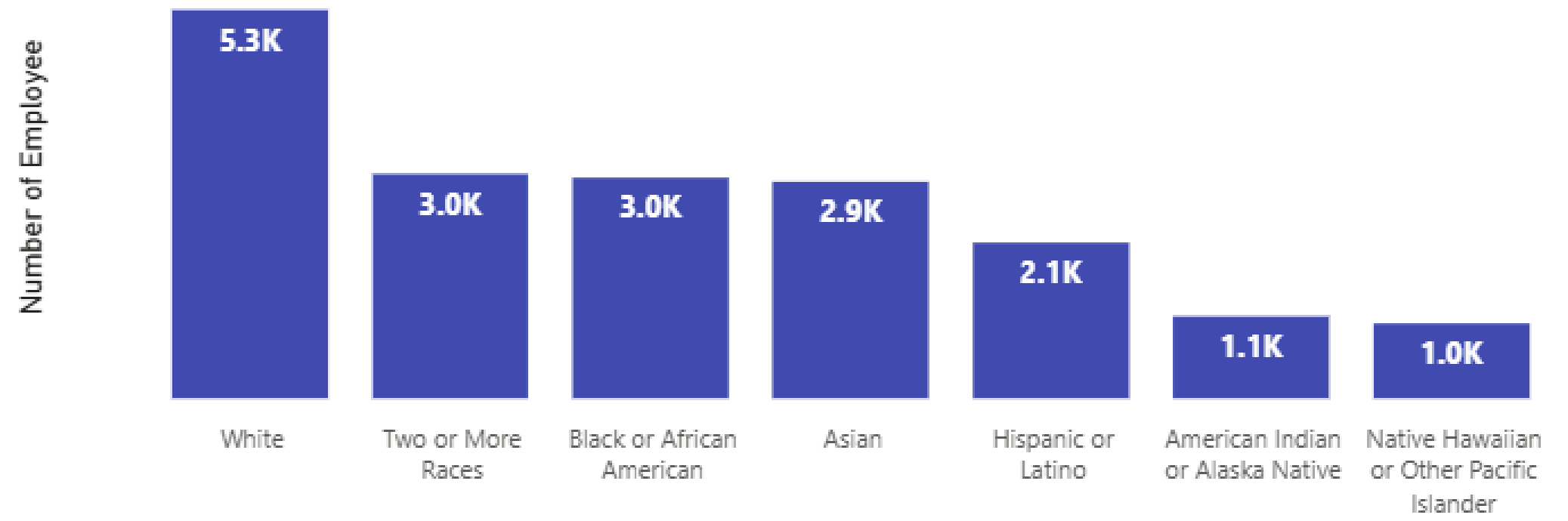
Change in Employee Numbers (2000-2020)



Employees by State



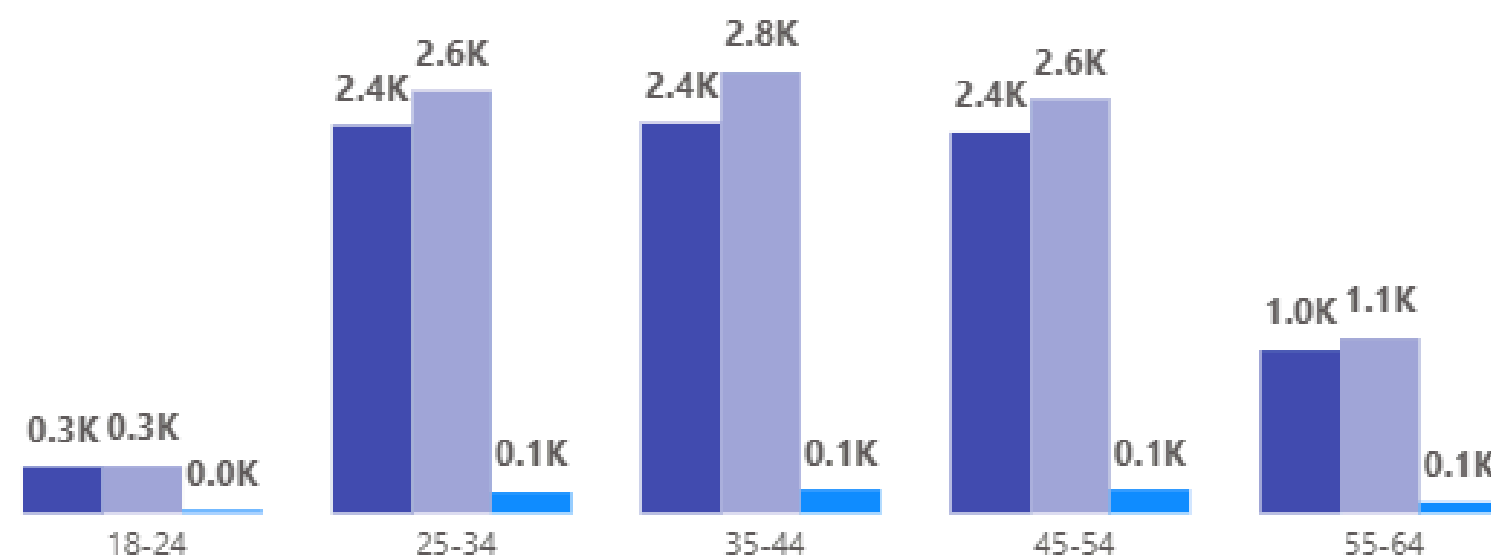
Race distribution



Age Distribution by Gender

gender ● Female ● Male ● Non-Conforming

Count of Employees



age_group

department

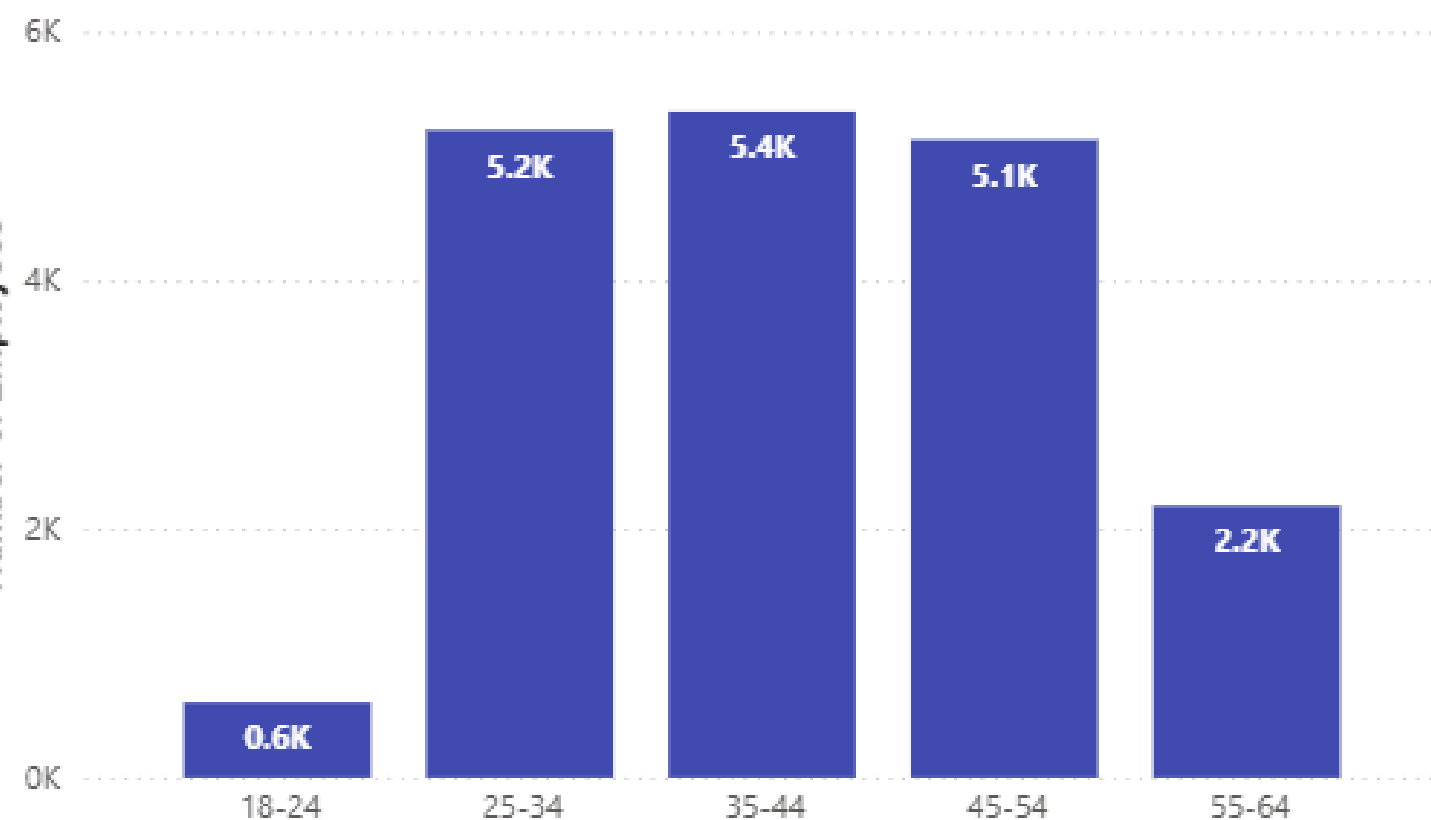
Sum of termination_rate

Auditing	0.20
Legal	0.17
Training	0.14
Support	0.14
Sales	0.13
Human Resources	0.13
Engineering	0.12

Gender Distribution by Department

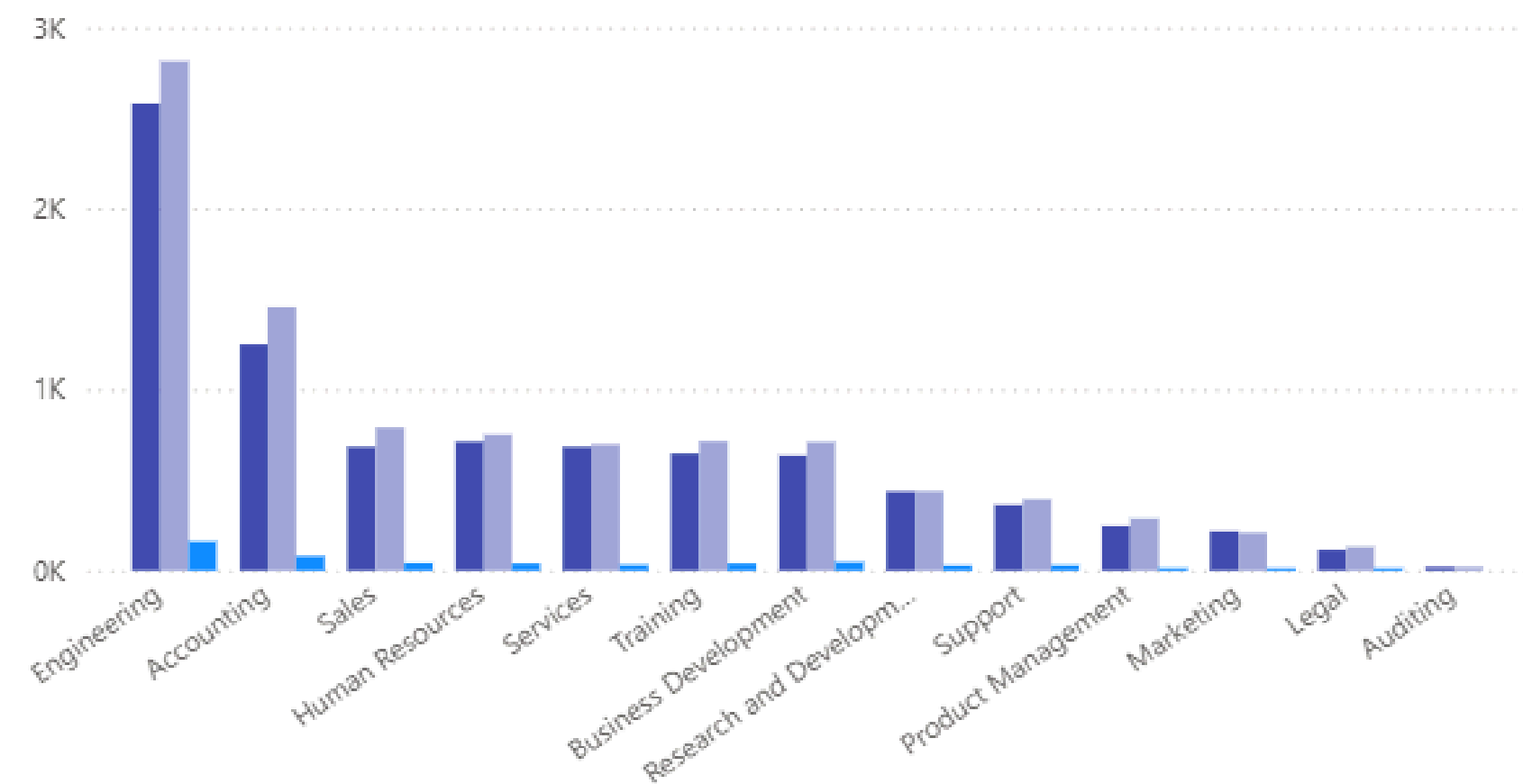
Age group distribution

Number of Employees



gender ● Female ● Male ● Non-Conforming

No. of Employees



VIEW POWER BI PROJECT →



SUMMARY OF FINDINGS

The HR Employee Distribution Report provides insights into the company's workforce demographics, employee distribution, and workforce trends. The analysis shows that the organization has a fairly balanced gender distribution, with male employees slightly outnumbering female employees, while non-conforming employees represent a very small proportion of the workforce.

The workforce is primarily composed of employees aged 25–54 years, with the 35–44 age group representing the largest segment. This indicates that the company has a mature and experienced workforce, while employees aged 18–24 and 55–64 make up smaller proportions.

Most employees work at the headquarters (approximately 75%), while around 25% work remotely, suggesting that the organization mainly operates through on-site employees.

From a diversity perspective, White employees form the largest racial group, followed by employees identifying as Two or More Races, Black or African American, and Asian. This reflects a workforce with multiple ethnic backgrounds.

The employee trend analysis shows a steady increase in workforce size between 2000 and 2020, indicating consistent hiring and organizational growth over time.

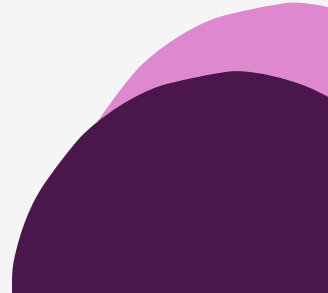
Departmental analysis reveals that Engineering employs the largest workforce, while Auditing records the highest employee turnover rate (20%), followed by Legal and Training. These departments may require additional attention to improve employee retention.

The average employment duration of terminated employees is 8 years, suggesting that employees generally remain with the company for a considerable period before leaving.

Overall, the report indicates a growing organization with a stable workforce, balanced gender representation, and opportunities to improve retention in departments experiencing higher turnover.



KEY INSIGHTS

- Male and female employee numbers are relatively balanced.
 - Employees aged 35–44 form the largest workforce segment.
 - Around 75% of employees work at headquarters.
 - Workforce size has grown steadily from 2000–2020.
 - Engineering has the highest number of employees.
 - Auditing has the highest turnover rate (20%).
 - Average employee tenure before termination is 8 years.
 - Workforce diversity is represented across multiple racial and ethnic groups.
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PROJECT FILES

FILE	DESCRIPTION	ACCESS
RAW DATA	Original Excel dataset	https://drive.google.com/file/d/1wBhJLIXBOovyjkjQdkNYtvEg4PAZ2HQB/view?usp=drive_link
SQL CLEANING	Data cleaning queries	https://drive.google.com/drive/folders/1BMDdqDYRkh7vYoLOX44epKPgu2V1z6v4?usp=drive_link
SQL QUERIES	Business analysis queries	https://drive.google.com/file/d/1_08c71z4VD9y_tV4N9bb3JawFqRYldj3/view?usp=sharing
POWER BI DASHBOARD	Interactive dashboard	https://drive.google.com/file/d/1Enu5a856s8-khEn0cu9ZAxq0MzYmiaZ9/view?usp=sharing
GITHUB	Interactive dashboard	https://github.com/analystkavin/HR-Employee-Analytics-Project.git



Thank You